

CHIEH TSAI



Linkedin

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Core Skills & Expertise

Research area: Autonomous Driving | Robotics | Cyber Physical Systems | AI

Programming: C Programming | Python | Verilog | PyTorch | TensorFlow | Matlab | Simulink | ROS2

Engineering & Research Background

Autonomic Computing Lab – University of Arizona

Arizona, USA

Graduate Research Assistant, Department of ECE

Aug. 2024-Current

- The study aims to design resilience methods to handle malicious attacks on autonomous vehicles, including the data fusion layer, data processing layer, decision layer, and control layer.
- Various threat models were simulated and applied to autonomous vehicles.
- The resilience methods were designed, including the integration of LiDAR, Depth cameras, RGB camera, and software communication frameworks

Course ECE 274A – University of Arizona

Arizona, USA

Teaching Assistant, Department of ECE

Aug. 2024-Current

- Assisting students in a Verilog and Vivado course. The course covers fundamental topics, including basic digital logic, LED display control, finite state machines (FSMs), and RTL Design.

VIP Lab, National Taiwan University of Science Technology

Taipei, Taiwan

Graduate Research Assistant, Department of ECE

July. 2023-Aug 2024

- The proposed model eliminates the window attention mechanism, substituting it with spatial and frequency self-attention to reinforce superior super-resolution detail learning, thereby enhancing the model's ability to capture finer details through spatial and frequency enhancements.
- Evaluated on the FGSCR dataset, with a specific focus on ship images, the proposed model achieves a notable 0.51 PSNR, 0.24 SSIM improvement and a 7% reduction in GFLOPs compared to the baseline SwinIR model.

VIP Lab, National Chi Nan University

Taipei, Taiwan

Graduate Research Assistant, Department of EE

Sep. 2019-June 2019

- Worked with a team of 3 to design an AI-equipped self-driving vehicle for campus navigation and security systems; won 3rd place in the department's project showcase and a cash prize of NT\$3,000
- Spearheaded software integration, including image feature recognition such as gender and age; designed the software and hardware component integration and successfully integrated a self-driving car

Patents, Publications, and Presentations

1. RACF: A Resilient Autonomous Car Framework with Object Distance Correction, 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (Submission)
2. Aerial-borne Data Management Center (ADMC), AAAI 2026 Spring Symposium Series (SSS-26), (Accepted)
3. Security and Resilience of Autonomous Systems, in Connected, Cooperative and Autonomous Mobility – Research Needs, Challenges and Future Perspectives, ISBN: 978-1-83635-539-7, 2026. (Accepted for publication)
4. [Enhancing satellite image quality with the edge-based wavelet transformer for super-resolution](#), 2025, Available at SSRN 5459834, Journal of Applied Computing and Geosciences (Q1, CiteScore 2024: 4.6)
5. [Intelligent Obstacle Resilience in Autonomous Vehicles Under Security Threats](#), 2025 IEEE Cyber Security and Cloud Computing (Accepted, Acceptance rate 24% in 2024), 2025, New York, USA.
6. [Spatial and Wavelet Attention-Enhanced Super-Resolution for Small Object Detection in Satellite Imagery](#), IEEE 8th open international conference electrical, electronic and information sciences eStream 2024, Republic of Lithuania
7. [Enhancing Age and Gender Classification through cGAN-based Data Augmentation](#), IEEE Global Conference on Consumer Electronics (GCCE), Oct. 10-13, 2023, Nara, Japan
8. [Real-Time Semantic Segmentation with Dual Encoder and Self-Attention Mechanism for Autonomous Driving](#), Sensors, 2021, 21.23: 8072

Education

PhD in Electrical and Computer Engineering | University of Arizona (USA) |(GPA: 4.0) Sep. 2024-current (Advised by [Dr. Hariri Salim](#)) [Cloud and Autonomic Computing Center](#) Funded by NSF and UA TRIF

MS in Electronic and Computer Engineering | National Taiwan University of Science Technology (Taiwan) |(GPA: 3.7) Sep. 2020-June 2023